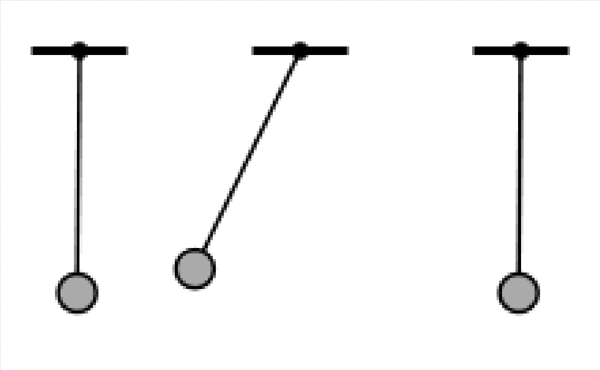
Day 24 - Driven Oscillations

Resonance in a driven pendulum system. ⟶

Source: Wikipedia



Announcements

- Midterm 1 is still being graded (sorry, bit off more than I can chew)
- Homework 6 is due Friday
- Homework 7 is posted, due next Friday
    - Final project check-in #1
- **Friday's Class**: We will complete HW 6 Exercises 2 and 4 together.

Seminars this week

TUESDAY, October 21, 2025

Theory Seminar, 11:00am., FRIB 1200 lab, In person and online via Zoom

Speaker: Antonio Bjelcic, LLNL

Title: Small and Large Amplitude Collective Dynamics within Nuclear Density Functional Theory

Zoom Link: 964 7281 4717

Meeting ID: 48824

Seminars this week

WEDNESDAY, October 22, 2025

FRIB Nuclear Science Seminar, 3:30pm., FRIB 1300 Auditorium and online via Zoom

Speaker: Dr. Nan Liu of Boston University

Title: Presolar Grains as Probes of Type II Supernova Nucleosynthesis

Please click the link below to join the webinar:

Join Zoom Meeting: https://msu.zoom.us/j/93944167137?pwd=jzvwvbL8YqDnJNpzDPat8IHcrFdtC5.1

Meeting ID: 939 4416 7137

Passcode: 239049

Seminars this week

THURSDAY, October 23, 2025

Colloquium, 3:30 pm, 1415 BPS, in person and zoom. Host ~

Refreshments and social half-hour in BPS 1400 starting at 3 pm

Speaker: Darryl Seligman, MSU

Title: Divergent Small Bodies: Interstellar Interlopers and Dark Comets

Zoom Link: https://msu.zoom.us/j/94951062663

Password: 2002 Or complete link: https://msu.zoom.us/j/94951062663?pwd=c48uM25P9UsRVuR74rkOioOWgpoxgC.1

Seminars this week

FRIDAY, October 24, 2025

QuIC Seminar, 12:30pm, -1:30pm, 1300 BPS, Virtual only today

Speaker: James Whitefield, Dartmouth

Title: TBA

Full Schedule is at: https://sites.google.com/msu.edu/quic-seminar/

Seminars this week

FRIDAY, October 24, 2025

IReNA Online Seminar, 2:00 pm, In Person and Zoom, FRIB 2025 Nuclear Conference Room, Light refreshments will be served at 1:50pm.

Hosted by: Hosted by: Linda Lombardo (INAF Trieste)

Speaker: Marta Molero, TU Darmstadt

Title: Chemical evolution of neutron-capture elements: a multi-objective approach

Zoom Link: https://msu.zoom.us/j/827950260

Password: JINA

Reminders

We solved the damped harmonic oscillator equation:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$$

We chose a solution (**ansatz**) of the form

$$x(t) = C_1 e^{rt} + C_2 e^{rt}$$

and computed the roots of the characteristic equation:

$$r^2 + 2\beta r + \omega_0^2 = 0$$

We found the roots to be:

$$r = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$$

Weak Damping

We found that when $\beta^2 < \omega_0^2$, the roots are complex:

$$r = -\beta \pm i\sqrt{\omega_0^2 - \beta^2}$$

This means that the solution is oscillatory:

$$x(t) = e^{-\beta t}\left(C_1 \cos(\sqrt{\omega_0^2 - \beta^2}t) + C_2 \sin(\sqrt{\omega_0^2 - \beta^2}t)\right)$$

The solution is a damped oscillation with frequency $\omega_1 = \sqrt{\omega_0^2 - \beta^2}$.

Strong Damping

When $\beta^2 > \omega_0^2$, the roots are real:

$$r = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$$

This means that the solution is not oscillatory:

$$x(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

where $r_1 = -\beta + \sqrt{\beta^2 - \omega_0^2} < 0$ and $r_2 = -\beta - \sqrt{\beta^2 - \omega_0^2} < 0$.

The solution is the sum of two exponentials with different decay rates.

Critical Damping

When $\beta^2 = \omega_0^2$, the roots are real and equal (repeated roots):

$$r = -\beta$$

This means that the solution is not oscillatory, but also that our ansatz is not sufficient. The correct form of the solution is:

$$x(t) = (C_1 + C_2 t)e^{-\beta t}$$

In most cases, we will work with weak damping.

Clicker Question 24-1

How are you doing?

1. Semester is going well; I'm locked in.
2. Semester is going alright; Tough but I'm working on it.
3. Semester is not going great TBH; I need to make some changes.
4. Danny, this semester is some BS.

Clicker Question 24-2

What do we expect the phase space diagram (x vs $\dot{x}$) to look like for a weakly damped harmonic oscillator?

1. A set of ellipses
2. A set of spirals
3. Depends on how weak the damping is
4. Depends on the total energy
5. More than one of the above

Clicker Question 24-3

The driven harmonic oscillator equation is:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f(t)$$

with $w_0^2 = k/m$ and $2\beta = b/m$. What is the dimension of the driving force $f(t)$?

1. Force (Newtons, N)
2. Force per unit second (N/s)
3. Force per unit length (N/m)
4. Force per unit mass (N/kg)

Clicker Question 24-4

The driven harmonic oscillator equation is:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f(t).$$

This ODE is a ________ differential equation.

1. linear
2. nonlinear
3. first-order
4. second-order
5. more than one of the above

Example: Sinusoidal Driving Force

Let $f(t) = f_0 \cos(\omega t)$, so that the driven harmonic oscillator equation is:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f_0 \cos(\omega t)$$

Note: $\omega \neq \omega_0$

Note that if the driving follows a sine wave, then we have:

$$\ddot{y} + 2\beta\dot{y} + \omega_0^2 y = f_0 \sin(\omega t)$$

Interesting, $e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$, let try to work with $z(t) = x(t) + iy(t)$.

Clicker Question 24-5

We found that the square amplitude of the driven harmonic oscillator is:

$$A^2 = \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

When is the amplitude of the driven oscillator maximized?

1. When the driving frequency (ω) is far from the natural frequency (ω_0)
2. When the driving frequency (ω) is close to the natural frequency (ω_0)
3. When the damping (2β) is weak
4. When the damping (2β) is strong
5. Some combination of the above